

$\Delta(\text{ICOBI}_{\text{antibonding}})$: A Fast Bond-Order-Based Descriptor for the Kinetic Viability of Metastable Crystalline Compounds

Gilles Frapper*

*IC2MP, UMR 7285 CNRS, Université de Poitiers, 4 rue Michel Brunet, TSA 51106,
86073 Poitiers Cedex 9, France*

E-mail: gilles.frapper@univ-poitiers.fr

Abstract

Crystal structure prediction routinely generates far more thermodynamically metastable candidates than the phonon, elastic-constant, and *ab initio* molecular dynamics calculations needed for a full stability audit can accommodate. We propose $\Delta(\text{ICOBI}_{\text{antibonding}})$, the change in the near-Fermi-level antibonding population of the Crystal Orbital Bond Index upon a compound’s decomposition into its elements, as an inexpensive, single-static-calculation pre-screen for the kinetic viability of such candidates, and test it, alongside its ICOHP-based counterpart, on a Materials-Project-derived dataset of 597 unary and binary structures (438–439 decomposition reactions) spanning stable, experimentally realized metastable, and theoretical-only metastable compounds. $\Delta(\text{ICOBI}_{\text{antibonding}})$ correlates significantly with formation energy ($\rho = -0.429$, $n = 438$), significantly discriminates experimentally realized from theoretical-only compounds (Fisher’s exact test, $p = 2.4 \times 10^{-7}$), and, unlike every other ICOHP/ICOBI-derived descriptor tested — including the fully integrated reaction $\Delta(\text{ICOHP})$ and the raw, non-reaction antibonding population — survives stratification by bonding type (metallic, ionic, covalent, mixed) in every subgroup large enough to test. Because it requires no calculation beyond the single static DFT+LOBSTER run already used to characterize a compound’s bonding, $\Delta(\text{ICOBI}_{\text{antibonding}})$ offers a practical route to concentrating the far more expensive stages of a stability hierarchy on the metastable candidates most likely to correspond to isolable compounds.

Introduction

Crystal structure prediction (CSP) — and, in the molecular realm, the closely related problem of conformer/isomer generation — has become a routine step in the computational discovery of new inorganic materials. Evolutionary and random-sampling algorithms such as USPEX,¹ CALYPSO,² AIRSS,³ and XtalOpt⁴ now explore compositional and configurational space largely unassisted, generating, for a single target composition, dozens to hundreds of candidate polymorphs, and, across a chemical system, an even larger set of

allotropes and stoichiometries.⁵ The traditional deliverable of such a search is an energetic ranking: candidate structures are relaxed at the density-functional-theory (DFT) level and ordered by total energy (or enthalpy, under pressure) at 0 K, and, for a multi-component system, plotted against composition to construct the thermodynamic convex hull — the set of compounds for which no combination of competing phases (elements or other compounds of the same system) offers a lower-energy alternative at that composition.^{6,7} Structures lying on the hull are, by definition, thermodynamically stable at 0 K and $P = 0$; every other stationary point found by the search is *metastable*, sitting some energy ΔE_{hull} above it.

Being a stationary point of the potential energy surface (PES) — a local minimum surviving geometry relaxation — is a necessary but far from sufficient condition for a predicted compound to correspond to a real, isolable material. A complete assessment of stability is, at minimum, hierarchical over several physically distinct criteria: (i) *energetic* stability, position relative to the convex hull, already discussed above; (ii) *dynamical* stability, the absence of imaginary phonon frequencies throughout the Brillouin zone, which rules out spontaneous symmetry-breaking distortions and requires a full lattice-dynamics calculation;⁸ (iii) *mechanical/elastic* stability, the positive-definiteness of the elastic constant tensor (the Born–Huang criteria), obtained from a set of finite strains and stress-tensor evaluations;⁹ (iv) *thermal/kinetic* stability at finite temperature, typically probed by *ab initio* molecular dynamics (AIMD) runs long enough to detect decomposition, amorphization, or a phase transition;¹⁰ and, for open or multinary systems, (v) *chemical* stability against exchange with an external reservoir, governed by the elemental chemical potentials and requiring, in principle, its own grand-canonical convex-hull construction. These five criteria are not redundant — a compound can sit exactly on the convex hull and still be dynamically or mechanically unstable, or conversely persist as a long-lived metastable phase despite a positive ΔE_{hull} — but they are also not equally affordable. Energetic ranking requires one static/relaxation calculation per candidate; phonon and elastic-constant calculations require, respectively, a supercell (or density-functional perturbation theory) treatment and a handful of strained-

cell single points; AIMD trajectories long enough to be diagnostic typically cost two to three orders of magnitude more CPU time than a single static calculation. In a CSP workflow that routinely produces 10^2 – 10^4 candidate structures per target composition, applying the full hierarchy to every metastable candidate is computationally prohibitive, and in practice only a small, pre-selected subset — usually the lowest-energy candidates — ever reaches the phonon or AIMD stage.

This asymmetry between the number of thermodynamically metastable stationary points a CSP search returns and the number that can be subjected to a full stability audit motivates the search for cheap, single-point descriptors able to pre-screen metastable candidates for their likelihood of being experimentally realizable, deferring the expensive criteria above to a much smaller shortlist. The conceptual basis for such a pre-screen is not new: it is what Hoffmann and coworkers have termed the *viability* of a predicted compound — the question of whether a computed stationary point, however far above the convex hull, corresponds to a structure that can actually be synthesized and persist, as opposed to one that will spontaneously collapse back to its elements or to a lower-lying phase the moment a synthesis is attempted.^{11,12} Viability, in this sense, is fundamentally a kinetic property — it concerns the existence (or absence) of an activation barrier protecting the candidate structure from decomposition — rather than a thermodynamic one, and it is precisely the criterion that energetic ranking and convex-hull position cannot, on their own, resolve: many experimentally well-characterized compounds sit measurably above the hull of their own chemical system, while many hull-adjacent theoretical structures have never been made despite repeated attempts.

A chemical-bonding-analysis route to this question can be built on the Crystal Orbital Hamilton Population (COHP), which partitions the band-structure energy into pairwise bonding ($\text{COHP} < 0$) and antibonding ($\text{COHP} > 0$) contributions, and whose energy integral up to the Fermi level, the ICOHP, quantifies the net strength of a given bond.^{13,14} Summing ICOHP values over every bond broken and formed in the decomposition of a com-

pound into its constituent elements yields a reaction descriptor, $\Delta(\text{ICOHP})$, whose sign we use below to classify a compound’s decomposition as *endobondic* (bond-breaking dominates, a kinetic barrier to decomposition is plausible) or *exobondic* (no such barrier is visible from bonding alone); this bonding-based reasoning is attractive precisely because it is affordable: unlike phonons or AIMD, it requires only a single self-consistent DFT calculation per structure, post-processed with LOBSTER,^{14,15} the program that projects plane-wave/PAW wavefunctions onto a local atomic-orbital basis to recover COHP and the closely related Crystal Orbital Bond Index (COBI/ICOBI),^{16,17} a dimensionless bond-order index that follows the opposite sign convention (positive = bonding).

The present work builds on this bonding-based reasoning and asks a more targeted question: rather than integrating the whole occupied COHP/COBI trace, does restricting the integration to the antibonding population confined to a narrow energy window just below the Fermi level (or the valence-band maximum, for gapped compounds) — the region most directly implicated in Peierls- and Jahn–Teller-type electronic instabilities — yield a sharper, more broadly applicable viability descriptor? We construct such a quantity for ICOBI, $\Delta(\text{ICOBI}_{\text{antibonding}})$, defined as the change, upon decomposition of a compound into its elements, of the near-frontier antibonding COBI population, and we validate it against a cohort of several hundred compounds spanning the full spectrum from thermodynamically stable, through experimentally realized but metastable, to theoretical (never-synthesized) metastable structures, drawn from the Materials Project.⁷ The remainder of this article is organized as follows: Section details the dataset, the DFT+LOBSTER computational protocol, and the precise definitions and sign conventions used for every ICOHP/ICOBI-derived quantity; Sections and (Results and Discussion) present the correlation of $\Delta(\text{ICOBI}_{\text{antibonding}})$ — and, for comparison, of the parent ICOHP-based descriptors — against thermodynamic stability targets and against a binary experimental/theoretical viability label, stratified by bonding character; Section concludes.

Methodology

Dataset

Candidate compounds were drawn from the Materials Project database⁷ by querying unary and binary chemical systems, restricted to non-magnetic structures (absolute total magnetization $\leq 0.01 \mu_B$ per formula unit — computed directly from the stored magnetic moments rather than from the Materials Project’s categorical `magnetic_ordering` field, which is unsigned for the large majority of entries) and excluding *f*-block elements. The final dataset combines three complementary selection tracks totaling 597 structures: (i) a first, broadly representative set (20 atoms/cell or fewer, balanced across stability classes, capped at two entries per chemical system to maximize elemental diversity over redundant polymorphs); (ii) a second set (40 atoms/cell or fewer) specifically targeting additional elemental references, off-equilibrium polymorphs (high-pressure phases, allotropes), and alkali/alkaline-earth binaries against N, O, F, P, S, and Cl, chosen to build genuine same-composition polymorph groups and to probe descriptor behavior far from equilibrium; and (iii) a third, deliberately adversarial set of 45 binary compounds pairing an experimentally known compound sitting farthest above the convex hull of its chemical system with one to three theoretical polymorphs of the same formula lying even farther above the hull, targeting the stability edge cases where a viability descriptor is most needed and most likely to fail. Each entry carries the Materials Project stability label (on-hull, experimentally observed metastable, or theoretical-only metastable, the last serving as the closest available proxy for a compound that has been searched for but never made) and, where available, `formation_energy_per_atom` and `energy_above_hull` as continuous thermodynamic targets.

Electronic-Structure and Bonding-Analysis Protocol

All structures were treated with an identical computational template to avoid introducing system-dependent methodological bias. Static (single-point) DFT calculations were

performed with the Vienna Ab initio Simulation Package (VASP),¹⁸ using the projector augmented-wave (PAW) method^{19,20} with the PAW-PBE potentials recommended by `pymatgen`’s `MPRelaxSet`, the PBE generalized-gradient exchange–correlation functional,²¹ symmetry explicitly disabled (`ISYM=−1`) to avoid artifacts in the subsequent bonding projection, and non-spherical gradient corrections enabled (`LASPH=.TRUE.`). The number of bands was set, per compound, from the number of local orbitals required by the LOBSTER basis rather than from a fixed multiplier, and basis-function assignment for each species was obtained programmatically rather than by parsing POTCAR headers directly, to avoid failures on potentials not recognized by hash in the local mirror. Chemical-bonding analysis was then performed with LOBSTER 5.1.1,^{14,15} projecting the converged VASP wavefunction onto the `pbeVaspFit2015` local basis, with a common COHP/COBI integration window from $−35$ to $+5$ eV around the Fermi level, bond detection by interatomic distance from 0.1 to 6.0 Å (orbital-resolved), and 0.05 eV Gaussian smearing. This identical template was applied to every structure in the dataset (Section details the internal quality-control diagnostics used to screen the resulting ICOHP/ICOBI output before any descriptor was computed from it).

ICOHP and ICOBI: Definitions and Sign Conventions

The Crystal Orbital Hamilton Population, $\text{COHP}(E)$, resolves the band-structure energy of a pair of atoms into an energy-dependent bonding (negative) and antibonding (positive) contribution.¹³ Its integral up to the Fermi level, the ICOHP, is by convention negative for a net bonding interaction and positive for a net antibonding one, and is reported in eV. The Crystal Orbital Bond Index, $\text{COBI}(E)$, and its integral ICOBI, are constructed analogously but derived directly from the density matrix rather than from the band-structure energy, and quantify covalent bond order rather than bond strength;¹⁶ critically, ICOBI follows the *opposite* sign convention to ICOHP — positive ICOBI denotes a net bonding interaction, as for the older Crystal Orbital Overlap Population (COOP) — so that the antibonding contribution to ICOBI is its *negative*, not positive, part. This distinction,

verified numerically against `ICOBILIST.lobster` output, is essential whenever ICOHP- and ICOBI-derived descriptors are combined or compared, and every quantity defined below states explicitly which convention it follows.

Two extensive per-compound aggregates are used as a baseline throughout this work: `icohp_sum/icobi_sum`, summing every symmetrically inequivalent bond label reported by LOBSTER within its detection window (each periodic bond counted once, irrespective of the number of equivalent directions), and the corresponding intensive per-bond averages, `icohp_mean/icobi_mean`. No coordination-shell filtering is applied to these baseline aggregates, so that every compound in the dataset is treated identically regardless of its local coordination chemistry.

Reaction $\Delta(\text{ICOHP})$ and the Endobondic/Exobondic Classification

For the decomposition of a compound into its constituent elements (“case-1” reactions in the sense of Bartel *et al.*,²² i.e. reactions whose products are exclusively elemental references), the reaction descriptor is defined, following the products–reactants thermochemical convention, as

$$\Delta(\text{ICOHP}) = \sum_{\text{products}} \nu_i \text{ICOHP}_i - \sum_{\text{reactants}} \nu_j \text{ICOHP}_j, \quad (1)$$

with ν the stoichiometric coefficients of the balanced reaction, computed in three normalizations (per formula unit, per transferred atom — the normalization used throughout unless stated otherwise — and, diagnostically only, per bond, which is not strictly conserved between differently-bonded structures). A positive $\Delta(\text{ICOHP})$ (*endobondic*) indicates that breaking the compound’s bonds costs more, in ICOHP terms, than the bonds of the free elements recover — a signature compatible with a kinetic barrier protecting the compound from decomposition; a negative value (*exobondic*) indicates no such barrier is visible from bonding alone. The sign of $\Delta(\text{ICOHP})$ is combined with the sign of the thermodynamic driving force for decomposition (ΔE , from `formation_energy_per_atom`) to assign each

compound a three-way viability label: a compound is classified `UNSTABLE_NONEXISTENT` only if decomposition into elements is *both* exobondic *and* thermodynamically favorable; a compound with $\Delta E \geq 0$ or an endobondic decomposition is classified `STABLE_ON_HULL` or `METASTABLE_VIABLE` as appropriate. This combined label, rather than the sign of $\Delta(\text{ICOHP})$ in isolation, is the target used to test experimental/theoretical discrimination in Section .

Near-Frontier Antibonding Population and Its Reaction Δ

A distinct construction targets not the total bonding integrated over the whole occupied manifold, but specifically the antibonding population confined to a narrow window adjacent to the frontier orbitals, by analogy with electronically driven (Peierls/Jahn–Teller-type) instabilities, for which only occupied antibonding character near E_F is expected to be destabilizing. For ICOHP, this quantity integrates only the antibonding part of the COHP trace ($\text{COHP} > 0$) over a one-sided energy window $(E_{\text{ref}} - \Delta E, E_{\text{ref}}]$, where E_{ref} is the Fermi level for metallic compounds (`is_metal = true`) or the valence-band maximum for gapped compounds, and $\Delta E \in \{0.5, 1.0, 2.0\}$ eV, with $\Delta E = 1.0$ eV as the primary window reported here. The identical construction is applied to the COBI trace, reading from `COBICAR.lobster/ICOBILIST.lobster` instead of `COHPCAR.lobster/ICOHPLIST.lobster`; because ICOBI’s sign convention is the mirror image of ICOHP’s (Section), the antibonding part of ICOBI is taken as its *negative* region, not its positive one — a deliberate mirror-image truncation, not a copy of the ICOHP procedure. Both quantities are computed from LOBSTER’s own pre-aggregated, per-bond-weighted “average” COHP/COBI trace rather than from a sum over discrete bond pairs, so that, unlike the aggregates of Section , no coordination-shell or bond-pair filtering enters their definition; only the energy window near E_{ref} does. Both are reported in raw units (eV for the ICOHP-based quantity, dimensionless for the ICOBI-based one, though neither is literally an energy or a bond order once restricted to a sub-window) and normalized by the total occupied ICOHP (respectively ICOBI) magnitude at the same E_{ref} .

Because this near-frontier antibonding population is, by construction, an intensive quantity (a per-bond average over the whole occupied trace, with no natural extensive decomposition into individual bonds), its reaction counterpart cannot be built by the same stoichiometric summation as Eq. 1. Instead, for a compound decomposing into elements k with atomic fractions x_k ,

$$\Delta(\text{antibonding}) = \sum_k x_k \text{antibonding}_k - \text{antibonding}_{\text{compound}}, \quad (2)$$

again in the products–reactants convention, with $\text{antibonding}_k/\text{antibonding}_{\text{compound}}$ evaluated either on the ICOHP-based or the ICOBI-based near-frontier population, yielding two descriptors, $\Delta(\text{ICOHP}_{\text{antibonding}})$ and $\Delta(\text{ICOBI}_{\text{antibonding}})$ respectively; the latter is the central descriptor of this work. A compound whose formation *reduces* near-frontier antibonding character relative to its constituent elements corresponds to $\Delta(\text{antibonding}) > 0$.

Bonding-Type Classification

Because the strength and interpretation of ICOHP/ICOBI-derived quantities depend on bonding character, every compound and element was additionally assigned to one of five categories — metallic, ionic, covalent, mixed (Zintl-type), or unclassified — used throughout as a stratification variable rather than as an end in itself. Two independent classification routes were compared. The first is a composition-based heuristic using only elemental identity and the `is_metal` flag: `is_metal` compounds with ≥ 2 elements and no anion-forming element present (halogens, O, N, S, Se, Te, H, C, P, As, B) are labeled metallic; among the remainder, an alkali/alkaline-earth element co-occurring with a halogen or O gives ionic, and a covalent-forming element (B, C, Si, Ge, N, P, As, Al, Ga, In, Sn, Pb, S, Se, Te) with no alkali/alkaline-earth present gives covalent; all other compositions are left unclassified. The second, used for every stratified result reported below, is built directly from the LOBSTER output of Section : `is_metal` again gives metallic; otherwise the compound’s

dominant species pair (highest mean ICOBI over its first coordination shell, among all pairs LOBSTER reports) is compared against a threshold calibrated as the midpoint between the median dominant-pair ICOBI of compounds the first (composition-based) route already labels ionic and those it labels covalent (0.5315 in the current dataset), giving ionic below and covalent at or above that value; a compound whose *homoatomic* pair (e.g. N–N, P–P, S–S) reaches the threshold while a distinct, weaker *heteroatomic* pair is also present — the Zintl–Klemm signature of a covalent polyanion loosely bound to a separate cation — is assigned to the mixed category instead, since forcing a single ionic/covalent label would misrepresent it. First coordination shells, where used (this classification and the validation reactions of Section only), are delimited by an adaptive, self-calibrating rule on the sorted bond-distance spectrum of each species pair (cutting at the first gap exceeding $2.5\times$ the largest gap seen so far within the putative shell), rather than by a fixed distance cutoff.

Statistical Analysis

Because no linear relationship is assumed *a priori* between any ICOHP/ICOBI-derived descriptor and a thermodynamic stability target, association is quantified throughout by Spearman rank correlation (ρ) against `formation_energy_per_atom` and `energy_above_hull`, computed both over the full dataset and within each bonding-type/metallicity subgroup of Section ; any subgroup with fewer than 15 members is reported but explicitly flagged as underpowered, never presented without that caveat. Discrimination between experimentally realized and theoretical-only (never synthesized) compounds is tested with Fisher’s exact test on the corresponding 2×2 contingency table (endobondic/exobondic, or descriptor sign, versus experimental/theoretical status) and, for the underlying continuous descriptor, with a Mann–Whitney U test; the same pair of tests is applied to the combined `ViabilityLabel` of Section against experimental/theoretical status, and to $\Delta(\text{ICOHP}_{\text{antibonding}})/\Delta(\text{ICOBI}_{\text{antibonding}})$ against a binary viable/non-viable split derived from the same label.

Quality Control

Before any descriptor was computed, every LOBSTER run was screened with two internal diagnostics available from its own output: the fraction of k -points for which the projected local basis fails to reach orthonormalization (`lobsterout`) and the maximum deviation reported in `bandOverlaps.lobster`, an established red-flag threshold of 0.1 being used to separate a numerically trustworthy projection from a pathological one. A systematic sweep of the full dataset under a conservative combined criterion (100% k -point failure *and* a band-overlap deviation > 0.5 *and* a robust z -score > 10 on the resulting $\Delta(\text{ICOHP})$) is reported in Section .

Results and Discussion

Dataset and Bonding-Type Classification

The final dataset (Section) comprises 597 VASP+LOBSTER calculations, of which 588 yield a usable ICOHP/ICOB_I record after quality control (Section); 439 of these participate in at least one decomposition-into-elements (case-1) reaction and therefore carry a $\Delta(\text{ICOHP})/\Delta(\text{antibonding})$ value. The bonding-based classification of Section assigns 372 compounds/elements as metallic, 98 as ionic, 56 as covalent, and 62 as mixed (Zintl-type), with every entry receiving a verdict. Where the composition-based heuristic also renders a verdict, the two classifications agree in 276/311 cases (88.7%); the disagreements concentrate in 19 compounds the heuristic calls ionic or covalent but whose dominant homoatomic bond (e.g. a polyanionic N–N, P–P, or S–S contact) the bonding-based route instead flags as mixed, and in a further 16 ionic/covalent swaps. Because the bonding-based classification also resolves the 277 compounds (47%) the composition-only heuristic leaves unclassified — chiefly alkali/alkaline-earth pairs with N, P, or S not covered by a fixed anion list — it is used throughout as the stratification variable below.

Baseline Correlations

Before turning to the near-frontier antibonding descriptors that are the focus of this work, three simpler ICOHP/ICOB-derivative quantities set a baseline. The classic per-compound aggregates (`icohp_sum`, `icohp_min`) correlate weakly or not at all with E_{hull} (Spearman $\rho = 0.032$ and 0.093 , $n = 186$, both non-significant; a weak but significant signal appears only in the metallic subgroup, $\rho = 0.223$, $p = 0.037$). The reaction descriptor $\Delta(\text{ICOHP})$ (Eq. 1) does better: pooled over 510–514 case-1 reactions, $\rho = 0.295$ against `formation_energy_per_atom` ($p \approx 0$) but only $\rho = -0.056$ ($p = 0.20$, not significant) against `energy_above_hull`. The endobondic/exobondic sign of $\Delta(\text{ICOHP})$ alone discriminates experimentally realized from theoretical-only compounds (Fisher’s exact test, odds ratio 1.81, $p = 0.0054$, endobondic decompositions being $\sim 1.8\times$ more frequent among experimentally realized compounds); combining that sign with the thermodynamic driving force into the three-way `ViabilityLabel` of Section sharpens this considerably (odds ratio 6.0, $\chi^2 = 59.0$, $p = 1.6 \times 10^{-14}$ on the collapsed viable/not-viable split) — of 188 experimentally realized compounds whose decomposition is exobondic, only 26 are in fact predicted non-viable, because for the other 162 that decomposition is thermodynamically unfavorable despite lacking a bonding-based barrier. This large gap between the sign-only test and the combined label is itself instructive: sign alone is a real but partial signal, and it is the combination with thermodynamics that carries most of the discriminating power in the ICOHP-integrated picture.

The raw near-frontier antibonding population (Section , not yet a reaction quantity) correlates with `formation_energy_per_atom` at $\rho = -0.155$ ($p = 1.8 \times 10^{-4}$, $n = 579$) for its ICOHP-based version — a counter-intuitive sign, since more occupied antibonding character near E_{F} is here associated with a *more*, not less, stable formation — and at $\rho = +0.231$ ($n = 579$) for its ICOB-derivative version, i.e. the naively expected direction. Table 1 compares the two, overall and by bonding-type subgroup.

Two observations motivate restricting attention, in what follows, to the *reaction* Δ of

Table 1: Spearman correlations of the raw near-frontier antibonding population (not a reaction quantity) against formation_energy_per_atom, ICOHP- versus ICOBI-based, same underlying population ($n = 588$; $\Delta E = 1.0$ eV window).

Subgroup	n	ρ (ICOHP)	n	ρ (ICOBI)
all	579	-0.155^{***}	579	$+0.231^{***}$
covalent	54	-0.046	54	$+0.113$
ionic	96	-0.196	96	$+0.162$
metallic	367	$+0.056$	367	$+0.120^*$
mixed	62	-0.538^{***}	62	$+0.025$

* $p < 0.05$; *** $p < 0.001$; no star = not significant.

this antibonding population rather than its raw value. First, the ICOHP- and ICOBI-based versions of the raw population run in *opposite* directions overall, a direct consequence of their opposite sign convention (Section) rather than of any disagreement in the underlying physics. Second, only the mixed (Zintl-type) subgroup survives stratification for either quantity at this scale; a stratified result reported for the covalent subgroup at a smaller sample size in earlier stages of this project did not survive once the sample grew to $n = 54$ ($\rho = -0.046$, $p = 0.74$), underscoring that any single-stratum result must be re-checked as a dataset grows rather than taken as settled.

$\Delta(\text{Antibonding})$ of the Decomposition Reaction

For a case-1 reaction, $\Delta(\text{antibonding})$ (Eq. 2) was computed for 438–439 reactions. Table 2 reports its correlation with both thermodynamic targets, for the ICOHP- and ICOBI-based versions side by side.

Table 2: Spearman correlations, $\Delta(\text{antibonding})$ of the decomposition-into-elements reaction ($n = 438\text{--}439$).

Descriptor	Target	n	ρ
$\Delta(\text{ICOHP}_{\text{antibonding}})$	formation energy/atom	438	-0.6225^{***}
$\Delta(\text{ICOBI}_{\text{antibonding}})$	formation energy/atom	438	-0.4292^{***}
$\Delta(\text{ICOHP}_{\text{antibonding}})$	energy above hull	438	-0.1734^{***}
$\Delta(\text{ICOBI}_{\text{antibonding}})$	energy above hull	438	-0.1215^*

* $p < 0.05$; *** $p < 0.001$.

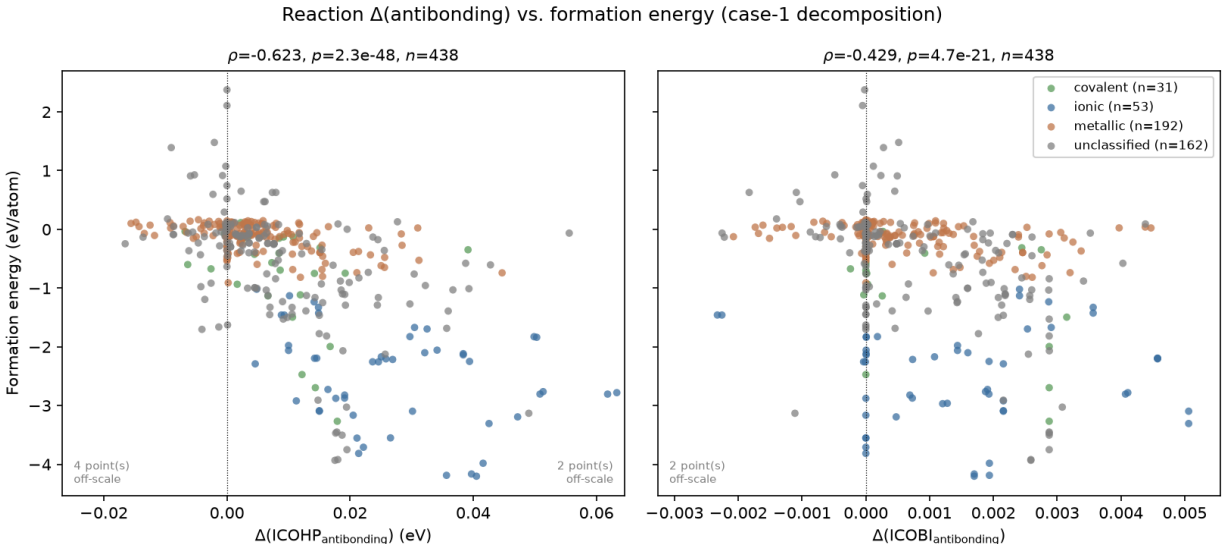


Figure 1: Formation energy per atom versus $\Delta(\text{ICOHP}_{\text{antibonding}})$ (left) and $\Delta(\text{ICOBI}_{\text{antibonding}})$ (right) of the decomposition-into-elements reaction, colored by bond type ($n = 438$). Spearman ρ and p (title of each panel) are computed on the full population; the x -axis is clipped to the 1st–99th percentile of each descriptor so that a small number of extreme-magnitude compounds do not compress the rest of the population, with the count of clipped points noted in-panel. The ionic subgroup (blue) drives most of the visible trend, consistent with Table 3.

Two features distinguish this reaction-level construction from the raw population of Section . First, $\Delta(\text{ICOHP}_{\text{antibonding}})$ is, at $\rho = -0.6225$, the strongest continuous correlation identified anywhere in this work — stronger than either the raw antibonding population or the fully integrated reaction $\Delta(\text{ICOHP})$ of Section ($\rho = 0.295$; note the reaction-ICOHP figure is reported in the products–reactants convention of `reaction_icohp.py`, opposite in sign to but equal in magnitude to the $\Delta(\text{ICOHP})$ convention of Eq. 1). Second, and more directly relevant to the present focus, $\Delta(\text{ICOBI}_{\text{antibonding}})$ — the central descriptor of this work — now runs in the *same* direction as its ICOHP counterpart ($\rho = -0.4292$, both negative), even though the raw, non-reaction antibonding populations they are built from run in opposite directions (Table 1). Taking the reaction Δ , i.e. referencing each compound’s near-frontier antibonding character against the atomic-fraction-weighted average of its own elements rather than reading it in isolation, thus removes the sign ambiguity between the ICOHP and ICOBI bond-order conventions and leaves a consistent, physically interpretable

picture: a compound whose formation reduces near-frontier antibonding character relative to its constituent elements ($\Delta > 0$ in our convention, i.e. a negative correlation with formation energy since more negative formation energies are more stable) is more likely to be thermodynamically favorable.

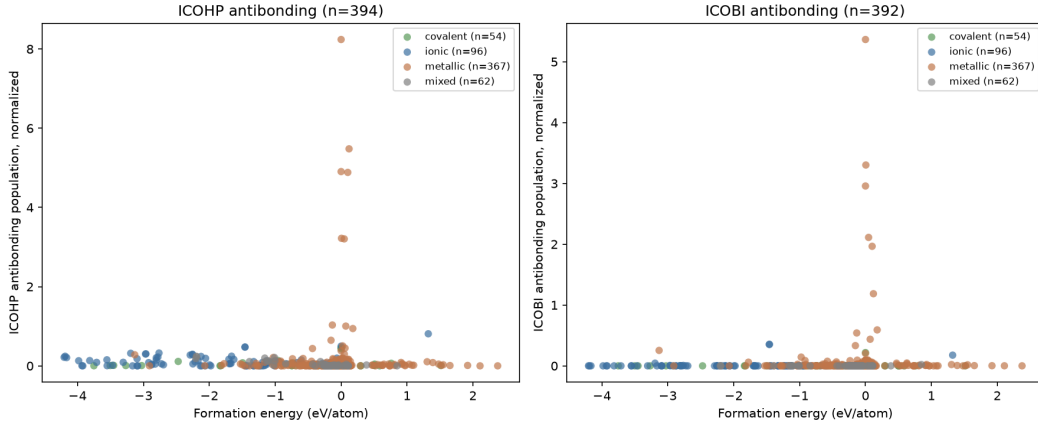


Figure 2: $\Delta(\text{ICOHP}_{\text{antibonding}})$ and $\Delta(\text{ICOB}_{\text{antibonding}})$ of the decomposition-into-elements reaction, both plotted against `formation_energy_per_atom` ($n = 438$).

Stratification by Bonding Type

Table 3 tests whether the pooled correlations above survive once conditioned on the bonding-type classification of Section .

Table 3: Spearman correlations, $\Delta(\text{antibonding})$ against `formation_energy_per_atom`, by bonding-type subgroup ($n = 438$ pooled).

Subgroup	n	$\rho (\Delta\text{ICOHP}_{\text{ab}})$	n	$\rho (\Delta\text{ICOB}_{\text{ab}})$
all	438	−0.6225***	438	−0.4292***
covalent	29	−0.6956***	29	−0.7750***
ionic	89	−0.6301***	89	−0.2258*
metallic	288	−0.4359***	288	−0.3602***
mixed	32	−0.6394***	32	−0.8397***
<code>is_metal=False</code>	150	−0.6466***	150	−0.4128***
<code>is_metal=True</code>	288	−0.4359***	288	−0.3602***

* $p < 0.05$; *** $p < 0.001$; no star = not significant.

This is, to our knowledge, the only result in this study — and, notably, the only one for

either the ICOHP- or the ICOBI-based version — that survives *every* stratification tested, covalent through metallic, `is_metal` on both sides, all significant at $p < 0.001$. By contrast, the raw antibonding population retains only two surviving strata (mixed, `is_metal=False`; Section) and the fully integrated reaction $\Delta(\text{ICOHP})$ survives on `bond_type` but not within the majority metallic stratum (Section). $\Delta(\text{ICOBI}_{\text{antibonding}})$ reaches its strongest single-stratum value on the mixed (Zintl-type) subgroup ($\rho = -0.840$), a category whose bonding — a strongly covalent homoatomic polyanion alongside a much weaker heteroatomic, more ionic contact — is precisely the kind of chemically heterogeneous bonding picture a single pooled descriptor is most likely to average away; that both $\Delta(\text{antibonding})$ quantities instead sharpen *within* this subgroup suggests the near-frontier restriction is tracking a genuinely local electronic feature rather than a composition-driven confound. The smaller subgroups here (covalent, $n = 29$; mixed, $n = 32$) fall below this project’s own $n = 15$ significance-reporting floor only in the sense of statistical power, not in significance itself — both remain significant at $p < 0.001$ — but should still be read as provisional pending a larger, independent sample.

Discrimination of Experimentally Realized from Theoretical-Only Compounds

The continuous correlations above test agreement with a thermodynamic target; a stricter and more directly relevant test is whether $\Delta(\text{antibonding})$ discriminates compounds that have actually been made from theoretical-only candidates that, despite comparable or lower E_{hull} , have never been synthesized. Using the same `ViabilityLabel` convention as Section (viable = `STABLE_ON_HULL` $\cup$ `METASTABLE_VIABLE`, not viable = `UNSTABLE_NONEXISTENT`), Table 4 reports Mann–Whitney U on the continuous $\Delta(\text{ICOHP}_{\text{antibonding}})$ and Fisher’s exact test on its sign, by subgroup ($n_{\text{total}} = 434$).

A compound whose formation reduces near-frontier antibonding character relative to its constituent elements is significantly more likely to be classified viable than one whose formation increases it — overall and within every subgroup large enough to test, including

Table 4: $\Delta(\text{ICOHP}_{\text{antibonding}})$ versus viability, by subgroup.

Subgroup	n viable	n not viable	Mann–Whitney p	Fisher OR	Fisher p
all	329	105	6.7×10^{-12} ***	3.49	2.4×10^{-7} ***
<code>bond_type=metallic</code>	125	65	0.0028**	2.20	0.021*
<code>is_metal=True</code>	187	99	6.7×10^{-5} ***	2.38	0.00099***
<code>is_metal=False</code>	142	6	0.012*	7.88	0.028*

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. Median $\Delta(\text{ICOHP}_{\text{antibonding}}) = 0.0067$ eV (viable) vs. 3.2×10^{-6} eV (not viable), all-subgroup. `bond_type=covalent` (3 non-viable) and `bond_type=ionic` (1 non-viable) are too small to test.

`is_metal=False` despite its thin non-viable class (only 6 compounds). Combined with the stratification result of Section , this gives two independent lines of evidence — a continuous thermodynamic correlate and a categorical viability test — pointing the same way: reduced near- E_F antibonding character upon formation tracks not only how thermodynamically favorable a compound’s formation is, but which compounds actually persist.

Discussion

Three findings, taken together, motivate treating $\Delta(\text{ICOBI}_{\text{antibonding}})$ — and its ICOHP counterpart, reported throughout for comparison — as a genuine viability pre-screen rather than a restatement of formation energy under a different name. First, restricting the antibonding integration to a narrow window adjacent to the Fermi level (or the valence-band maximum) and taking its reaction Δ consistently outperforms both the classic ICOHP aggregates and the fully integrated reaction $\Delta(\text{ICOHP})$ baseline (Section), and does so for both the ICOHP- and the ICOBI-based construction, despite their raw (non-reaction) versions running in formally opposite directions. Second, unlike every other descriptor tested in this work, the reaction $\Delta(\text{antibonding})$ construction survives stratification by bonding type in *every* subgroup large enough to test (Section) — metallic, ionic, covalent, and mixed alike — rather than being carried by one dominant chemical class or flipping sign across strata, as the fully integrated reaction $\Delta(\text{ICOHP})$ does (ionic positive, covalent/mixed negative;

Section). Third, it discriminates experimentally realized from theoretical-only compounds directly (Section), not merely via a correlation with a continuous thermodynamic proxy for that status.

This combination of properties is what makes $\Delta(\text{ICOBI}_{\text{antibonding}})$ attractive specifically as a pre-screen ahead of the phonon/elastic/AIMD stability hierarchy described in Section : it requires nothing beyond the single static DFT+LOBSTER calculation already described in Section — no supercell, no finite-strain set, no finite-temperature trajectory — and, being a bond-order-based rather than a total-energy-based quantity, it is available immediately alongside a compound’s ICOHP/ICOBI bonding characterization rather than requiring an additional calculation. For a CSP campaign generating far more metastable stationary points than can be individually subjected to phonon or AIMD screening, ranking candidates by $\Delta(\text{ICOBI}_{\text{antibonding}})$ offers a physically motivated way to concentrate that downstream budget on the candidates most likely to correspond to kinetically persistent, isolable compounds.

Some caveats temper this picture. The descriptor is validated here on a single dataset assembled specifically to span the stability spectrum (Section), including a deliberately adversarial subset of theoretical polymorphs selected to sit far above the convex hull; whether the viability discrimination of Section partly reflects that selection choice rather than being an independent confirmation of the underlying bonding-kinetics mechanism has not yet been fully disentangled, and is a direct target for the independent-dataset test proposed in Section . The **theoretical**-only status used to define “not viable” throughout is itself a proxy — the absence of a synthesis report, not a measured kinetic barrier or a phonon-instability verdict — and several of this project’s smaller bonding-type subgroups (Table 3) remain modest in size. Finally, because the fully integrated reaction $\Delta(\text{ICOHP})$ changes sign across bonding-type strata (Section), any future symbolic-regression or feature-selection search built on these descriptors should be run separately per bonding type rather than on the pooled population, where such a search would risk being actively misleading.

Current Limits

A systematic sweep of the dataset flagged a single compound (a copper–gold intermetallic) for exclusion from every statistic reported above: its LOBSTER projection fails k -point orthonormalization at 100% of sampled k -points, with a maximum `bandOverlaps.lobster` deviation of 11.4 against this project’s own 0.1 red-flag threshold (Section), and correspondingly implausible ICOHP values (third-neighbor contacts with positive ICOHP an order of magnitude too large to be physical). The next-worst case in the full 597-structure sweep is already $29\times$ less extreme and plausibly reflects genuine, dense d -electron-rich intermetallic chemistry rather than a projection failure, confirming the excluded compound as a unique quality-control case rather than a symptom of a broader problem; every statistic in this article has been computed on the remaining structures.

Beyond this single exclusion, three further limitations bound the scope of the present analysis. First, the `theoretical`-only proxy for non-viability, discussed above, is not a true dynamical-instability label; extending the dataset with explicit phonon calculations for a representative subset would allow $\Delta(\text{ICOBI}_{\text{antibonding}})$ to be tested against a genuinely independent stability criterion rather than against synthesis history alone. Second, per-subgroup sample size remains this project’s structural limit at every new stratification: the raw antibonding population’s covalent-subgroup result reported at $n = 33$ in an earlier, smaller version of this dataset ($\rho = -0.551$) did not survive once the sample grew to $n = 54$ ($\rho = -0.046$; Section), a concrete illustration that any single-stratum result should be re-checked as a dataset grows rather than accepted once and for all. Third, because reaction $\Delta(\text{ICOHP})$ ’s signal changes sign across bonding-type strata (Section), a pooled symbolic-regression search (e.g. SISSO) across the whole dataset is deferred in favor of a future per-bonding-type search, left to subsequent work.

Conclusion

Restricting the integration of the ICOHP and ICOBI bonding-population traces to a narrow, one-sided window adjacent to the Fermi level (or the valence-band maximum), and referencing the resulting near-frontier antibonding population against a compound’s own constituent elements through the reaction quantity $\Delta(\text{antibonding})$ (Eq. 2), yields the most robust viability descriptor identified in this work. $\Delta(\text{ICOBI}_{\text{antibonding}})$ correlates significantly with formation energy ($\rho = -0.429$, $n = 438$), survives stratification by bonding type in every subgroup tested — a property no other descriptor in this study shares, including the fully integrated reaction $\Delta(\text{ICOHP})$ and the raw (non-reaction) antibonding population — and, alongside its ICOHP-based counterpart, significantly discriminates experimentally realized from theoretical-only metastable compounds (Fisher’s exact test, $p = 2.4 \times 10^{-7}$, overall). Because it is obtained from the same single static DFT+LOBSTER calculation already required to characterize a compound’s bonding, it offers a computationally inexpensive pre-screen for kinetic viability, to be applied ahead of the far more costly phonon, elastic-constant, and AIMD stages of the stability hierarchy outlined in Section , concentrating that downstream budget on the metastable CSP candidates most likely to correspond to isolable compounds.

Three directions follow directly from the limitations noted in Section . First, reproducing the present correlations and stratified results on an independently assembled dataset, ideally one not built with the current dataset’s deliberately adversarial, far-above-hull theoretical subset, would distinguish a genuine bonding-kinetics mechanism from a residual selection effect. Second, testing $\Delta(\text{ICOBI}_{\text{antibonding}})$ against explicit phonon and elastic-constant calculations for a representative subset would replace the synthesis-history proxy used here with a directly computed dynamical criterion. Third, a bonding-type-resolved symbolic-regression search (e.g. SISSO), run separately per stratum rather than on the pooled population, could clarify whether $\Delta(\text{ICOBI}_{\text{antibonding}})$ can be combined with complementary descriptors — or refined into a still sharper one — within each chemical class; extending the underlying dataset

to ternary and higher-order chemistry, and to explicit pressure dependence in line with the high-pressure CSP context of Section , are natural companions to that effort.

Acknowledgement

Acknowledgements and funding sources to be added.

Data Availability. The dataset, computational scripts, and analysis code underlying this article are available in the project’s repository (URL to be finalized for publication).

Supporting Information Available

Dataset composition and selection criteria; complete list of the compounds and elements studied; VASP and LOBSTER input parameters (INCAR, POTCAR, and k -point grids); LOBSTER quality-control diagnostics for the full dataset; the complete list of case-1 decomposition-into-elements reactions with $\Delta(\text{ICOHP})$, $\Delta(\text{ICOHP}_{\text{antibonding}})$, and $\Delta(\text{ICOB}_{\text{antibonding}})$ values; endobondic and exobondic reaction subsets.

References

- (1) Oganov, A. R.; Glass, C. W. Crystal Structure Prediction Using Ab Initio Evolutionary Techniques: Principles and Applications. *J. Chem. Phys.* **2006**, *124*, 244704.
- (2) Wang, Y.; Lv, J.; Zhu, L.; Ma, Y. CALYPSO: A Method for Crystal Structure Prediction. *Comput. Phys. Commun.* **2012**, *183*, 2063–2070.
- (3) Pickard, C. J.; Needs, R. J. Ab Initio Random Structure Searching. *J. Phys.: Condens. Matter* **2011**, *23*, 053201.

- (4) Falls, Z.; Avery, P.; Wang, X.; Hilleke, K. P.; Zurek, E. The XtalOpt Evolutionary Algorithm for Crystal Structure Prediction. *J. Phys. Chem. C* **2021**, *125*, 1601–1620.
- (5) Oganov, A. R.; Pickard, C. J.; Zhu, Q.; Needs, R. J. Structure Prediction Drives Materials Discovery. *Nat. Rev. Mater.* **2019**, *4*, 331–348.
- (6) Ong, S. P.; Wang, L.; Kang, B.; Ceder, G. Li–Fe–P–O₂ Phase Diagram from First Principles Calculations. *Chem. Mater.* **2008**, *20*, 1798–1807.
- (7) Jain, A.; Ong, S. P.; Hautier, G.; Chen, W.; Richards, W. D.; Dacek, S.; Cholia, S.; Gunter, D.; Skinner, D.; Ceder, G. et al. Commentary: The Materials Project: A Materials Genome Approach to Accelerating Materials Innovation. *APL Mater.* **2013**, *1*, 011002.
- (8) Togo, A.; Tanaka, I. First Principles Phonon Calculations in Materials Science. *Scr. Mater.* **2015**, *108*, 1–5.
- (9) Mouhat, F.; Coudert, F.-X. Necessary and Sufficient Elastic Stability Conditions in Various Crystal Systems. *Phys. Rev. B* **2014**, *90*, 224104.
- (10) Cohen, A.; Gerber, R. B. Decomposition Dynamics of a New Noble-Gas Compound. *Molecules* **2025**, *30*, 4398.
- (11) Hoffmann, R. How Chemistry and Physics Meet in the Solid State. *Angew. Chem. Int. Ed. Engl.* **1987**, *26*, 846–878.
- (12) Grochala, W.; Hoffmann, R.; Feng, J.; Ashcroft, N. W. The Chemical Imagination at Work in Very Tight Places. *Angew. Chem. Int. Ed.* **2007**, *46*, 3620–3642.
- (13) Dronskowski, R.; Blöchl, P. E. Crystal Orbital Hamilton Populations (COHP): Energy-Resolved Visualization of Chemical Bonding in Solids Based on Density-Functional Calculations. *J. Phys. Chem.* **1993**, *97*, 8617–8624.

- (14) Maintz, S.; Deringer, V. L.; Tchougréeff, A. L.; Dronskowski, R. LOBSTER: A Tool to Extract Chemical Bonding from Plane-Wave Based DFT. *J. Comput. Chem.* **2016**, *37*, 1030–1035.
- (15) Nelson, R.; Ertural, C.; George, J.; Deringer, V. L.; Hautier, G.; Dronskowski, R. LOBSTER: Local Orbital Projections, Atomic Charges, and Chemical-Bonding Analysis from Projector-Augmented-Wave-Based Density-Functional Theory. *J. Comput. Chem.* **2020**, *41*, 1931–1940.
- (16) Müller, P. C.; Ertural, C.; Hempelmann, J.; Dronskowski, R. Crystal Orbital Bond Index: Covalent Bond Orders in Solids. *J. Phys. Chem. C* **2021**, *125*, 7959–7970.
- (17) George, J.; Petretto, G.; Naik, A.; Esters, M.; Jackson, A. J.; Nelson, R.; Dronskowski, R.; Rignanese, G.-M.; Hautier, G. Automated Bonding Analysis with Crystal Orbital Hamilton Populations. *ChemPlusChem* **2022**, *87*, e202200123.
- (18) Kresse, G.; Furthmüller, J. Efficient Iterative Schemes for Ab Initio Total-Energy Calculations Using a Plane-Wave Basis Set. *Phys. Rev. B* **1996**, *54*, 11169–11186.
- (19) Blöchl, P. E. Projector Augmented-Wave Method. *Phys. Rev. B* **1994**, *50*, 17953–17979.
- (20) Kresse, G.; Joubert, D. From Ultrasoft Pseudopotentials to the Projector Augmented-Wave Method. *Phys. Rev. B* **1999**, *59*, 1758–1775.
- (21) Perdew, J. P.; Burke, K.; Ernzerhof, M. Generalized Gradient Approximation Made Simple. *Phys. Rev. Lett.* **1996**, *77*, 3865–3868.
- (22) Bartel, C. J.; Weimer, A. W.; Lany, S.; Musgrave, C. B.; Holder, A. M. The Role of Decomposition Reactions in Assessing First-Principles Predictions of Solid Stability. *npj Comput. Mater.* **2019**, *5*, 4.

TOC Graphic

*Table of Contents graphic to be inserted —
schematic of the decomposition-into-elements
reaction and the near-Fermi-level antibonding
window used to build $\Delta(\text{ICOB})_{\text{antibonding}}$.*